

MCQ-5

Topic: Descriptive Statistics and Data Measures

Student Name: _____ Date: _____ Score: _____

Instructions: Select the correct option by filling or marking the checkbox ☐ next to it.

1. Which of the following is not a property of the arithmetic mean?
 - ☐ The sum of deviations from the mean is always zero
 - ☐ It is affected by extreme values
 - ☐ It can be calculated for open-end classes without any assumption
 - ☐ The mean of a constant is the constant itself
2. For the data: 4, 7, 8, 10, 11. The sample variance (using $n - 1$) is:
 - ☐ 6.5
 - ☐ 7.5
 - ☐ 8.0
 - ☐ 9.5
3. A set of 5 observations has mean 10 and variance 4. If each observation is multiplied by 3 and then 2 is added, the new standard deviation is:
 - ☐ 6
 - ☐ 12
 - ☐ 14
 - ☐ 18
4. If $\text{Cov}(X, Y) = 0$, which statement is always true?
 - ☐ X and Y are independent
 - ☐ X and Y are uncorrelated
 - ☐ X and Y have zero correlation and are independent
 - ☐ The correlation coefficient is undefined
5. A dataset has a mean of 62, a median of 58, and a standard deviation of 8. Pearson's second coefficient of skewness (median-based) is:
 - ☐ A. 0.50
 - ☐ B. -0.50
 - ☐ C. 1.50
 - ☐ D. -1.50
6. In a normal distribution with mean 70 and SD 8, the z-score of the value 58 is:
 - ☐ -1.5
 - ☐ -1.25
 - ☐ 1.5
 - ☐ 1.25

7. For a distribution, Mean = 50, Mode = 45, SD = 10. Pearson's first coefficient of skewness is:
- ☐ 0.5
 - ☐ -0.5
 - ☐ 1.5
 - ☐ -1.5
8. A distribution with kurtosis = 0 or 3 is called:
- ☐ Platykurtic
 - ☐ Leptokurtic
 - ☐ Mesokurtic
 - ☐ Hyperkurtic
9. The fourth central moment $\mu_4 = 243$ and variance $\sigma^2 = 9$. The coefficient of kurtosis β_2 is:
- ☐ 3
 - ☐ 9
 - ☐ 27
 - ☐ 81
10. For the ordered data: 12, 15, 18, 20, 22, 25, 28, 30, 32, 35. The 70th percentile is approximately:
- ☐ 28
 - ☐ 29
 - ☐ 30
 - ☐ 31
11. For the data: 5, 8, 12, 15, 18, 20, 25, 28, 30. Q_1 and Q_3 are:
- ☐ 10 and 25
 - ☐ 8 and 28
 - ☐ 12 and 25
 - ☐ 12 and 28
12. A student's marks are 85. Class mean = 70, SD = 10. His z-score is 1.5. Which statement is correct?
- ☐ He is 1.5 SD below the mean
 - ☐ Approximately 93.3% of students scored less than him (assuming normality)
 - ☐ His percentile rank is exactly 85
 - ☐ Both A and C
13. Review the following block of code:

```
import numpy as np
x = np.array([1, 2, 3])
y = np.array([4, 5, 6])
print(np.cov(x, y, ddof=0)[0,1])
```

The code output will produce:

- ☐ 1.0
- ☐ 0.666...

- ☐ 2.0
 - ☐ 0
14. Which measure of central tendency is most affected by extreme outliers?
- ☐ Mean
 - ☐ Median
 - ☐ Mode
 - ☐ Quartile
15. Covariance between two variables X and Y measures:
- ☐ How much one variable's variance is scaled compared to another
 - ☐ The strength and direction of the linear relationship on a standardized (-1 to 1) scale
 - ☐ Whether two variables tend to increase/decrease together (direction only, unscaled)
 - ☐ The central value shared by X and Y
16. Given the array `A = np.arange(24).reshape(2, 3, 4)`, what will `A[0, 0, 0]` return?
- ☐ 0
 - ☐ 1
 - ☐ 24
 - ☐ An error
17. Given a 3D array `A = np.arange(24).reshape(2, 3, 4)`. Which of the following slices will yield an output array with a shape of (2, 2)?
- ☐ `A[:, 1:, 2]`
 - ☐ `A[:, :2, :2]`
 - ☐ `A[0, :, 1:3]`
 - ☐ `A[:, 1:3, 1:3]`
18. If a constant value 'c' is added to every observation in a dataset, the variance:
- ☐ Increases by c
 - ☐ Increases by c^2
 - ☐ Remains unchanged
 - ☐ Becomes zero
19. If every observation of a dataset is multiplied by a constant k , the standard deviation becomes:
- ☐ $SD \times k$
 - ☐ $SD \times k^2$
 - ☐ SD / k
 - ☐ SD (unchanged)
20. A dataset has mean = 20 and standard deviation = 4. What is the coefficient of variation (CV)?
- ☐ 5%
 - ☐ 20%
 - ☐ 80%
 - ☐ 25%